

HOMEWORK 12 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

1. Convert the IVP

$$y'' + 4y = \sin t, \quad y(0) = 0, \quad y'(0) = 1$$

into a first-order system. **do not solve!**

2. Given the system of algebraic equations

$$x + 3y + z = 1$$

$$1 + z + 2y = 0$$

$$y + z + x = 1$$

- (a) Convert the system into a matrix equation $A\vec{v} = \vec{b}$, where $\vec{v} = [x \ y \ z]$ is the vector of unknowns.
- (b) Without finding solutions, perform a calculation that can determine if this matrix equation has a *unique* solution vector?
- (c) Use Gauss elimination to find the value or a general formula for the solution vector.
3. Solve the nonhomogeneous system

$$\mathbf{x}' = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^t \\ t \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

using the **variation of constants formula**.

4. Given is the second order differential equation

$$2\ddot{y}(t) + c\dot{y}(t) + 8y(t) = 0,$$

where c is a parameter.

- (i) Write the ODE as the first-order system of differential equations. ¹
- (ii) Compute the eigenvalues of the obtained system (they will depend on c).
- (iii) Find the range of values of c that makes the equilibrium at origin *stable* and that results in oscillating trajectories (spiral phase portrait).

Show all your work.

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¹Introduce variables $x_1 = y$ and $x_2 = \dot{y}$.